Mitosis

36A

Mitosis: occurs in somatic cells and produces identical cells. Exact chromosome duplication occurs, and twin daughter cells are formed. Mitosis has four phases: prophase, metaphase, anaphase and telophase (occurring in that order).

Cut out the box above and glue it in your book as your title and introduction. Cut out the diagrams (provided on sheet 36B) and the descriptions below of the phases of mitosis. Match the diagrams with the correct descriptions and stick them beside each other in your book. The summary goes at the end. (Hint: the summary will help you arrange and match the boxes).

When all chromatids are aligned at the 'equator', they begin to pull apart. The 'twins' will now go their separate ways, half to the north 'pole' and half to the south. The centromere goes first, as if it were being pulled by a magnet. The 'arms' of the chromatid follow.

The longest phase. At the beginning of mitosis, chromosomes become more compact (condense) and undergo replication. When they decondense, we find two replicas of what was originally one chromosome. These 'twins' are now referred to as *chromatids*. Chromatids are joined by a link called a *centromere* (sometimes called a 'kinetochore' in older text books.) The centromere 'pinches' the twins together, forming the 'arms' on the chromatids. The short arm is called the '*p*-arm' and the long arm is called the '*q*-arm'.

The final phase. Genetic material has migrated to the poles. A new nuclear envelope reforms around each daughter cell. During this phase, the cell begins to pinch itself at the 'equator' to form two smaller, new cells. This process is called *cleavage*. (From the word 'cleave', meaning 'to cut'.) Half of the original cytoplasm goes into each cell. It is important to include some of the original cytoplasm in the new cells because it contains many organelles responsible for ensuring the genetic material within the nucleus can be used properly. The genetic material (DNA) can be compared to an architect's blueprint, it must be followed exactly.

Spindles form. Spindles are made of protein fibres and microtubules. As the name suggests, microtubules are tiny, tube-like structures. The spindles form an oval structure, along which chromatids can align themselves as they migrate to the 'equator'.

Summary

Prophase: the earliest stage of nuclear division. Chromosomes condense, replicate and decondense.

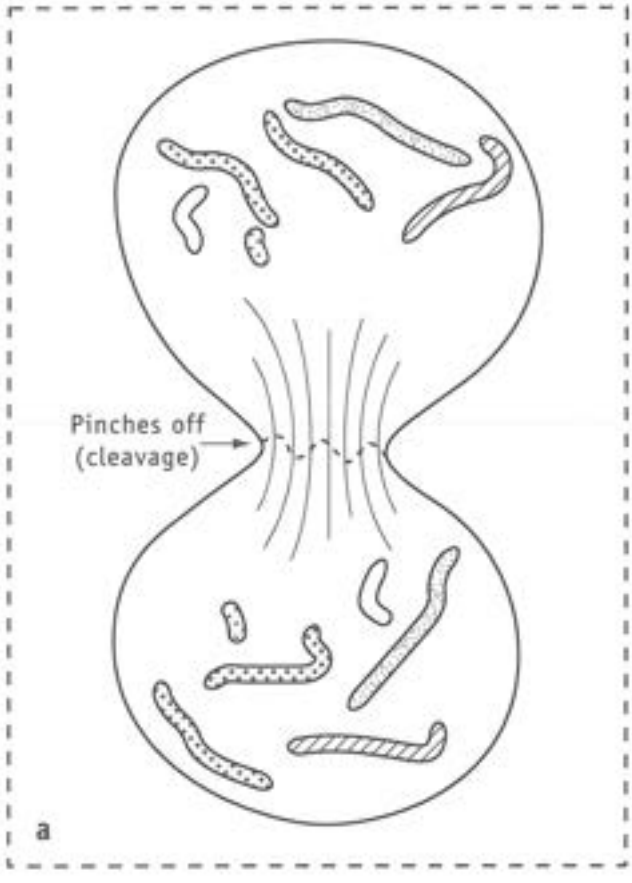
Metaphase: the chromosomes start aligning themselves along the 'equator' of the nucleus.

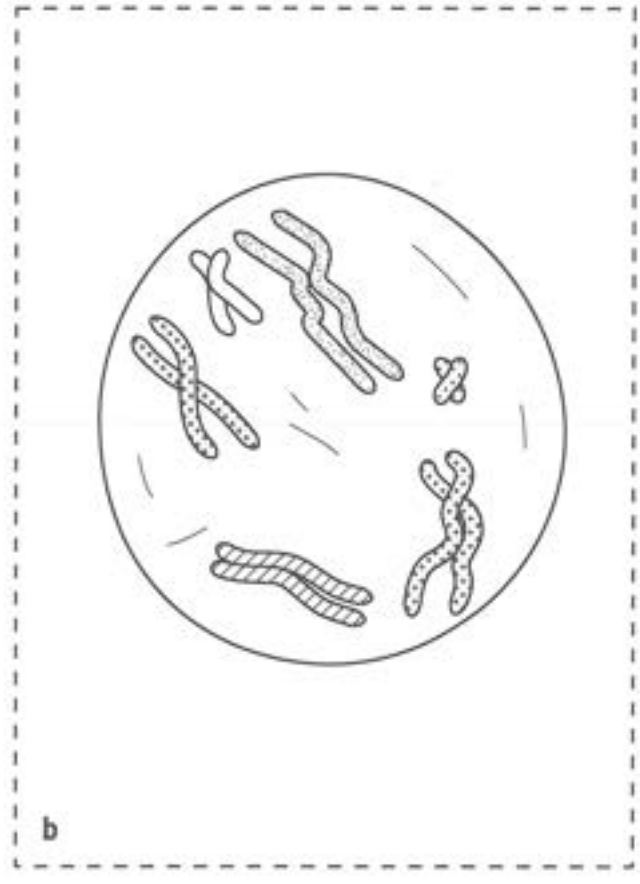
Anaphase: when chromosomes are aligned at the 'equator', they then move towards the poles.

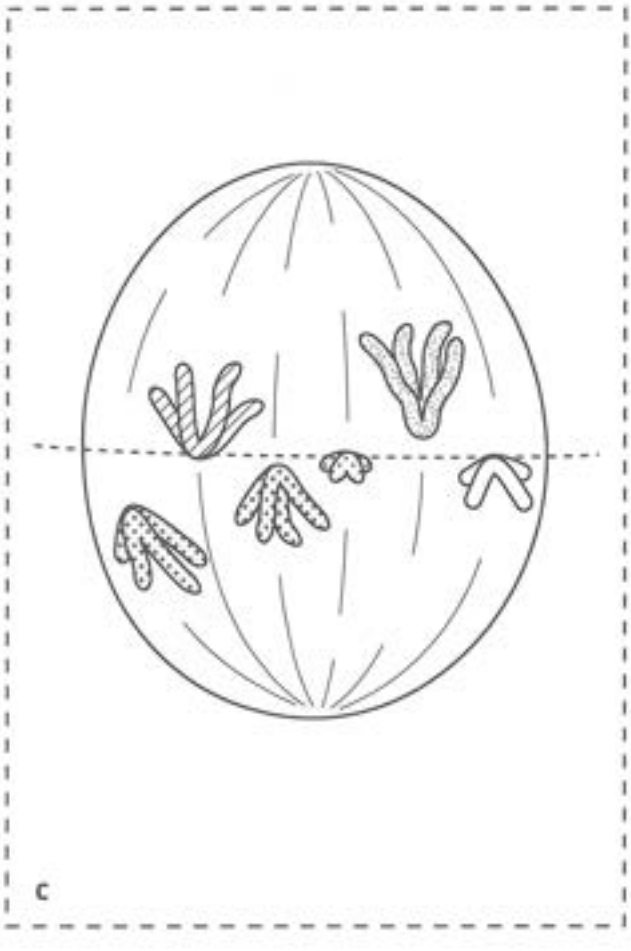
Telophase: the nuclear membrane reforms, enclosing the chromosomes in newly formed daughter cells.

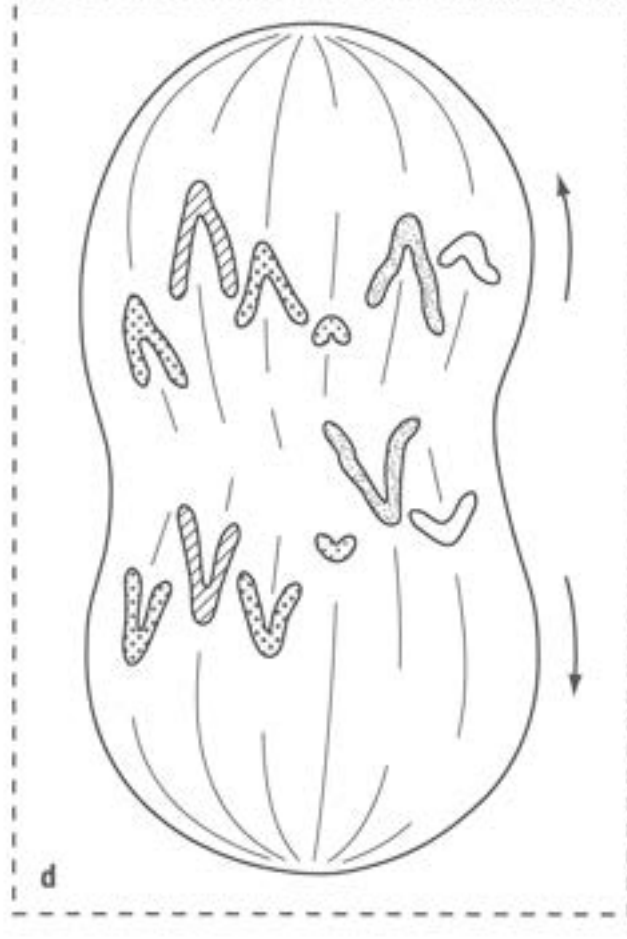
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